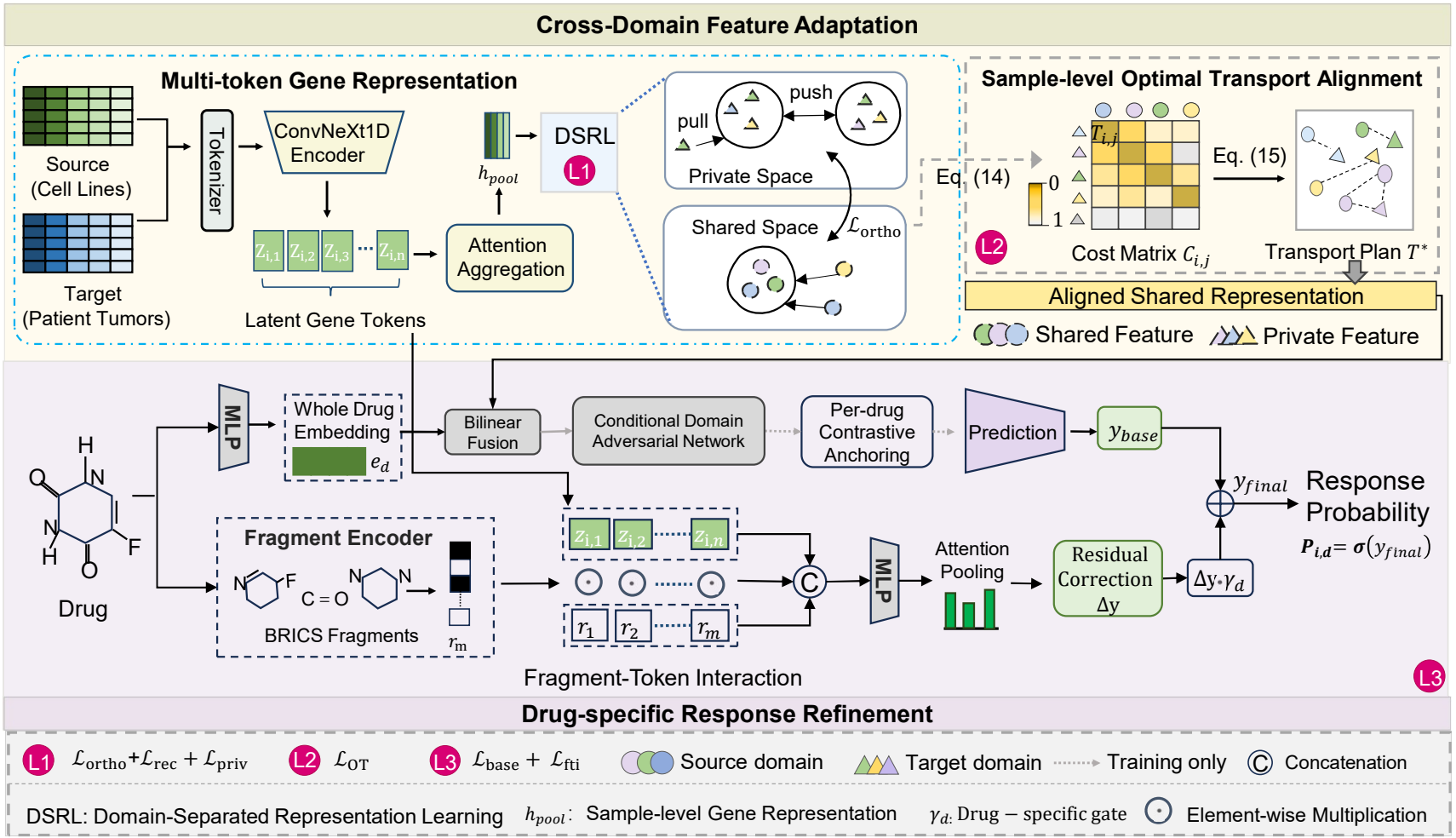

Cross-Domain Feature Adaptation
Multi-token Gene Representation
Source
(Cell Lines)
Target
(Patient Tumors)
Tokenizer
ConvNeXt1D
Encoder
Latent Gene Tokens
Attention
Aggregation
h_{pool}
DSRL
L1
pull
push
Private Space
Shared Space
$\mathcal{L}_{ortho}$
Sample-level Optimal Transport Alignment
Eq. (14)
$T_{i,j}$
0
1
Cost Matrix $C_{i,j}$
L2
Eq. (15)
Transport Plan T^*
Aligned Shared Representation
Shared Feature
Private Feature
Drug
MLP
Whole Drug
Embedding
e_d
Bilinear
Fusion
Conditional Domain
Adversarial Network
Per-drug
Contrastive
Anchoring
Prediction
y_{base}
Fragment Encoder
BRICS Fragments
r_m
C
MLP
Attention
Pooling
Residual
Correction
Δy
Δy·γ_d
y_{final}
Response
Probability
$P_{i,d}= \sigma(y_{final})$
Fragment-Token Interaction
L3
Drug-specific Response Refinement
L1 $\mathcal{L}_{ortho}+\mathcal{L}_{rec}+\mathcal{L}_{priv}$
L2 $\mathcal{L}_{OT}$
L3 $\mathcal{L}_{base}+\mathcal{L}_{fti}$
Source domain
Target domain
Training only
Concatenation
DSRL: Domain-Separated Representation Learning
h_{pool}: Sample-level Gene Representation
γ_d: Drug − specific gate
Element-wise Multiplication